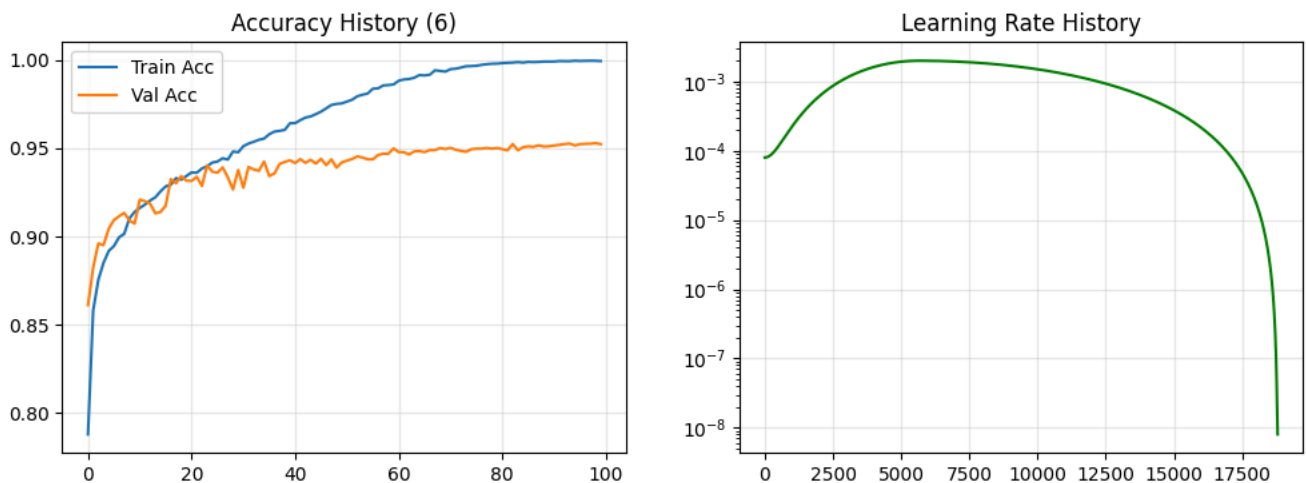


Pytorch 훈련 / 테스트 그래프 출력 화면

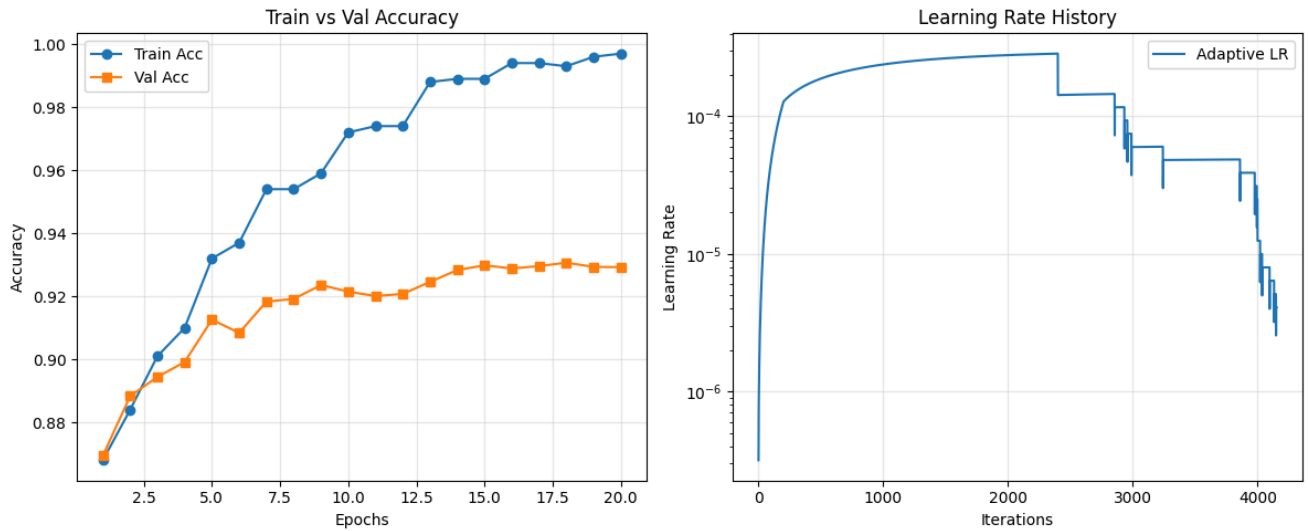


Pytorch log

```
Epoch 90/100 | Loss: 0.5159 | Train: 0.9990 | Val: 0.9511 | LR: 0.000099 | Time: 537.90s
Epoch 91/100 | Loss: 0.5155 | Train: 0.9990 | Val: 0.9515 | LR: 0.000080 | Time: 558.04s
Epoch 92/100 | Loss: 0.5151 | Train: 0.9993 | Val: 0.9518 | LR: 0.000064 | Time: 575.41s
  [Early Stopping] No improvement for 9 epochs.
Epoch 93/100 | Loss: 0.5151 | Train: 0.9992 | Val: 0.9523 | LR: 0.000049 | Time: 562.79s
Epoch 94/100 | Loss: 0.5151 | Train: 0.9992 | Val: 0.9526 | LR: 0.000036 | Time: 549.63s
  [Best] New Record! Val Acc: 0.9526 (Previous: 0.9523)
Epoch 95/100 | Loss: 0.5148 | Train: 0.9995 | Val: 0.9515 | LR: 0.000025 | Time: 480.49s
Epoch 96/100 | Loss: 0.5148 | Train: 0.9994 | Val: 0.9523 | LR: 0.000016 | Time: 499.52s
Epoch 97/100 | Loss: 0.5147 | Train: 0.9994 | Val: 0.9524 | LR: 0.000009 | Time: 516.23s
  [Early Stopping] No improvement for 3 epochs.
Epoch 98/100 | Loss: 0.5145 | Train: 0.9995 | Val: 0.9525 | LR: 0.000004 | Time: 524.97s
Epoch 99/100 | Loss: 0.5144 | Train: 0.9994 | Val: 0.9528 | LR: 0.000001 | Time: 524.83s
  [Best] New Record! Val Acc: 0.9528 (Previous: 0.9526)
Epoch 100/100 | Loss: 0.5150 | Train: 0.9993 | Val: 0.9523 | LR: 0.000000 | Time: 514.86s
Log ID 6 finalized and updated to 7.
```

```
=====
                        Training Finished
=====
flex_history_pt(6).pkl&flex_best_model_pt(6).pt
Final Val Acc (Best): 0.9528
Final Test Acc (Real): 0.9499
Total Elapsed Time: 15:00:39.560
```

Scratch 훈련 / 테스트 그래프 출력 화면



Scratch log

```
[Iter 100/200] Current Batch Loss: 0.0620
[Iter 150/200] Current Batch Loss: 0.0292
[Spike Brake] Iter 178: Loss 0.0878 튀어오름! LR 긴급 제어 1.9643901093064218e-05
[Spike Brake] Iter 195: Loss 0.0648 튀어오름! LR 긴급 제어 1.571508796239756e-05
[Iter 200/200] Current Batch Loss: 0.0297
[Dynamic Scheduler] Loss 정제 3회 발생! Base LR 50% 삭감 → 0.000013
Epoch 20/1000 | Loss: 0.0393 | TrainAcc: 0.9970 | ValAcc: 0.9293 | LR: 0.000025 | Time: 1961.96s

--- Epoch 21 / 1000 ---
[Spike Brake] Iter 21: Loss 0.0706 튀어오름! LR 긴급 제어 6.286231316559745e-06
[Spike Brake] Iter 37: Loss 0.0641 튀어오름! LR 긴급 제어 5.0290409566918035e-06
[Iter 050/200] Current Batch Loss: 0.0182
[Spike Brake] Iter 97: Loss 0.0662 튀어오름! LR 긴급 제어 4.023222651414953e-06
[Iter 100/200] Current Batch Loss: 0.0387
[Spike Brake] Iter 133: Loss 0.0520 튀어오름! LR 긴급 제어 3.218624577150322e-06
[Spike Brake] Iter 148: Loss 0.0555 튀어오름! LR 긴급 제어 2.574980073624737e-06
[floor] Iter 149: Loss 0.0441 but, lr이 floor에 도달했습니다.
[Iter 150/200] Current Batch Loss: 0.0441
[floor] Iter 150: Loss 0.0637 but, lr이 floor에 도달했습니다.
[floor] Iter 151: Loss 0.0359 but, lr이 floor에 도달했습니다.
[floor] Iter 152: Loss 0.0287 but, lr이 floor에 도달했습니다.
[floor] Iter 153: Loss 0.0455 but, lr이 floor에 도달했습니다.
[floor] Iter 153: Loss 0.0455 but, lr이 floor에 도달했습니다.

===== Training Finished =====

=====
Final Loss: 0.0393
Final Val Acc (Best): 0.9307
Final Test Acc (Real): 0.9260
```